

Bioinformatics session

4th annual workshop on
bioinformatics and variant
interpretation in InPreD

https://inpred.github.io/26-09_bioinfo_ws/



2. Containerization



A brief history

1979	<code>chroot</code> system call (changing the root directory of a process and its children to a new location in the filesystem) in Unix V7 considered as beginning of process isolation
2000	FreeBSD Jails allows administrators to partition FreeBSD computer system into several independent, smaller systems (<i>jails</i>) – with the ability to assign IP address for each system and configuration
2001	Linux VServer is a jail mechanism that can partition resources, similar to FreeBSD Jails

2004	First public beta of Solaris Containers released - combines system resource controls and boundary separation provided by zones
2005	Open Virtuozzo is operating system-level virtualization technology for Linux which uses a patched Linux kernel for virtualization, isolation, resource management and checkpointing
2006	Process Containers, launched by Google, was designed for limiting, accounting and isolating resource usage of a collection of processes - renamed to <i>Control Groups (cgroups)</i> and merged into Linux kernel
2008	Linux Containers (LXC) was the first, most complete implementation of a Linux container manager - implemented using cgroups and Linux namespaces

2011	Warden from Cloud Foundry can isolate environments on any operating system, runs as a daemon and provides API for container management
2013	Let Me Contain That For You (LMCTFY) kicked off as open-source version of Google's container stack - providing Linux application containers
2013	Docker emerged
2016	Singularity (later Apptainer) was released

Docker

What is it? 🔍

- a technology for packaging an application with all of its dependencies into an independent executable unit (a container)
- the container image can be shipped and run consistently across different computing environments

Which benefits does it provide?



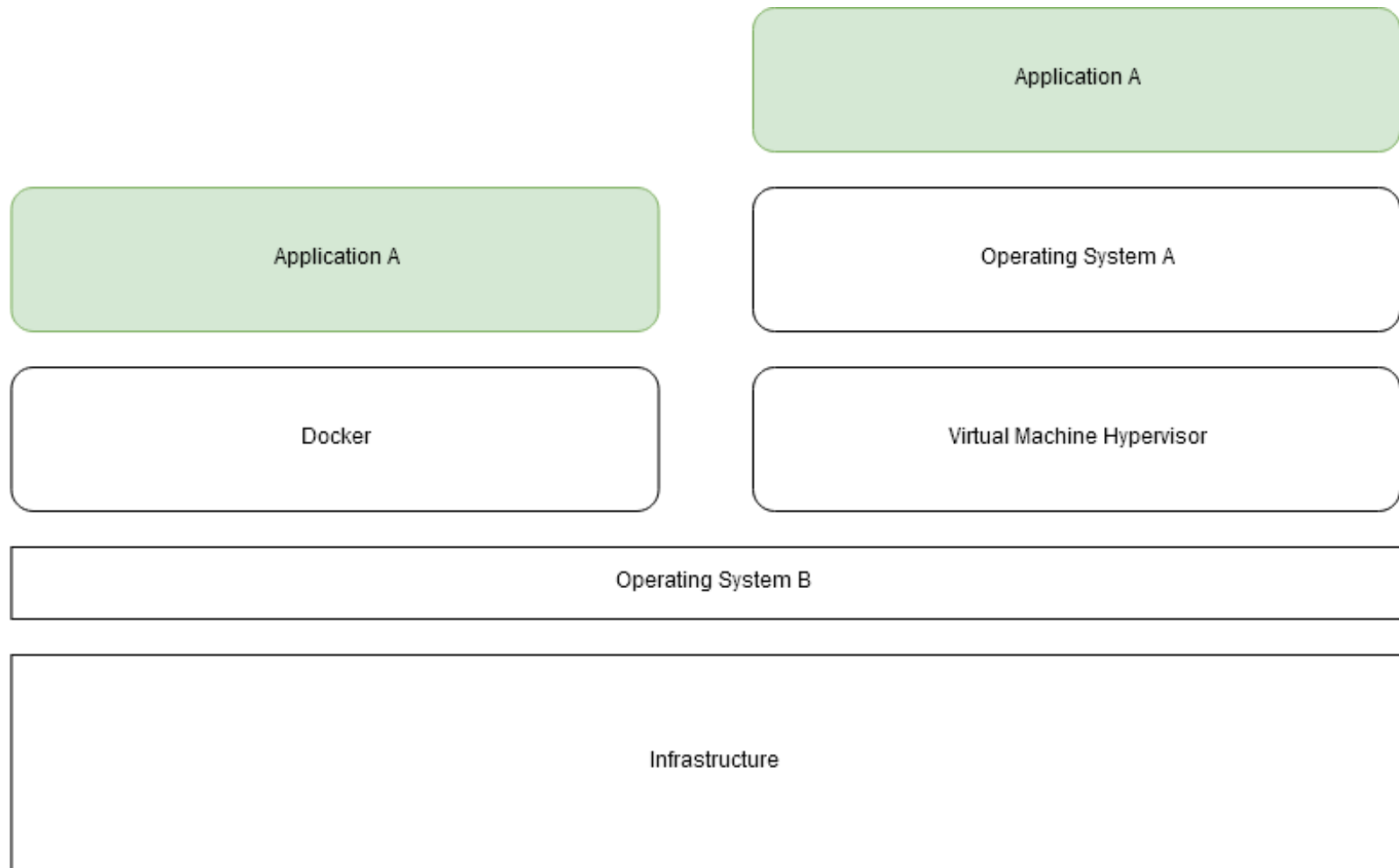
1. Consistency
2. Isolated environments
3. Portability
4. Efficiency

<https://scalablehuman.com/2023/10/24/the-fallacy-of-it-works-on-my-machine>



How is it different from virtual machines?

- containers use and share host OS's kernel, which makes them more lightweight and efficient
- virtual machines emulate entire physical machines, including the complete OS, making it possible to run multiple OS instances on a single physical machine



<https://dev.to/swikritit/docker-for-dummies-introduction-to-docker-5h67>

Key Concepts 🔑

1. **Image:** a standardized package that includes all files, binaries, libraries, and configurations necessary to run a given container
2. **Container:** a running instance of an image, operating in an isolated runtime environment
3. **Dockerfile:** instructions on how to build an image
4. **Docker Hub:** a public registry where developers can share their pre-built images

Key Components 🔑

1. **Docker Engine:** core, open-source technology that builds and runs containers
2. **Docker Client:** command-line tool that sends instructions to the Docker Daemon using REST APIs
3. **Docker Daemon (`dockerd`):** receives and processes API requests and calls container runtime
4. **Container Runtime (`containerd`):** turns static container image into running, isolated application on host OS; industry standard

Let's explore! 🌍

Start by going to https://github.com/InPreD/26-09_bioinfo_ws_docker_and_ci and create a fork.

The screenshot shows the GitHub interface for the repository `26-09_bioinfo_ws_docker_and_ci`, which is public. At the top, there are buttons for `Edit Pins`, `Watch` (0), `Fork` (0), and `Star` (0). Below these, a dropdown menu is open, displaying the message "You don't have any forks of this repository." and a button labeled `+ Create a new fork`, which is highlighted with a red rectangle. The repository is currently on the `main` branch, showing 1 branch and 0 tags. A search bar labeled "Go to file" and an `Add file` button are also visible. The file list includes `.devcontainer`, `src/greeter`, `.gitignore`, `LICENSE`, `README.md`, and `pyproject.toml`, each with its commit message and time. The `README.md` is selected, showing the title `26-09_bioinfo_ws_docker_and_ci` and the description `codespace template for docker and ci workshop`. On the right sidebar, there are sections for `workshop` (with links to `Readme`, `AGPL-3.0 license`, `Activity`, `Custom properties`, `0 stars`, `0 watching`, `0 forks`, `Audit log`, and `Report repository`), `Releases` (with `No releases published` and a link to `Create a new release`), and `Packages` (with `No packages published` and a link to `Publish your first package`).

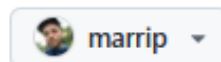
Create your own fork by clicking on **Create fork**.

Create a new fork

A *fork* is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project.

Required fields are marked with an asterisk ().*

Owner *



Repository name *

/ 26-09_bioinfo_ws_docker_

✓ 26-09_bioinfo_ws_docker_and_ci is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

codespace template for docker and ci workshop

45 / 350 characters

☒ Copy the `main` branch only

Contribute back to InPreD/26-09_bioinfo_ws_docker_and_ci by adding your own branch. [Learn more.](#)

You are creating a fork in your personal account.

Create fork

In the forked repository, navigate to **Code** > **Codespaces** > **Create codespace on main** .

The screenshot shows a GitHub repository page for '26-09_bioinfo_ws_docker_and_ci' (Public). The repository has 1 branch (main) and 0 tags. The 'Code' button is highlighted in green. A dropdown menu is open, showing 'Local' and 'Codespaces' options. The 'Codespaces' option is selected, and a sub-menu is displayed with the text 'No codespaces' and 'You don't have any codespaces with this repository checked out'. A green button labeled 'Create codespace on main' is highlighted with a red rectangle. Below this button is a link 'Learn more about codespaces...'. The repository description is 'codespace template for docker and ci workshop'. The repository has 0 stars, 0 watching, and 0 forks. The 'About' section shows the license as AGPL-3.0. The 'Releases' section shows 'No releases published' with a link 'Create a new release'. The 'Packages' section shows 'No packages published' with a link 'Publish your first package'.

26-09_bioinfo_ws_docker_and_ci Public

Edit Pins Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file T Add file <> Code

Local Codespaces

Codespaces
Your workspaces in the cloud

No codespaces
You don't have any codespaces with this repository checked out

Create codespace on main

Learn more about codespaces...

Codespace usage for this repository is paid for by marrip.

26-09_bioinfo_ws_docker_and_ci

codespace template for docker and ci workshop

About
codespace template for docker and ci workshop
Readme
AGPL-3.0 license
Activity
Custom properties
0 stars
0 watching
0 forks
Audit log
Report repository

Releases
No releases published
Create a new release

Packages
No packages published
Publish your first package

Inside the terminal, run the following commands:

```
# check for running docker containers
$ docker ps
# check for existing images
$ docker images
# pull image
$ docker pull ubuntu:26.04
# run image
$ docker run ubuntu:26.04
```

Run an interactive container:

```
# start interactive container
$ docker run -it ubuntu:26.04
# print container os version
@ cat /etc/lsb-release
# exit container
@ exit
# print os version
$ lsb_release -a
```

Instruct docker to remove containers after use:

```
# check for any docker containers
$ docker ps -a
# remove all stopped containers
$ docker rm <CONTAINER ID>
# run docker with --rm flag
$ docker run -it --rm ubuntu:26.04
# exit container
@ exit
# check for all docker containers
$ docker ps -a
```

Remove the docker image:

```
# remove docker image
$ docker rmi ubuntu:26.04
# alternatively check for container image id
$ docker images
# and remove the image by using the id
$ docker rmi <ID>
```

Building an image

We start off by creating a `Dockerfile` in the root directory of our repository (`/workspaces/26-09_bioinfo_ws_docker_and_ci`). We add the following to our file:

```
FROM python:3.14-slim-trixie
RUN echo "Hello world!" > greetings.txt
```

And then we build and run our docker image:

```
# build tagged docker image
$ docker build . -t greeter:test
# run tagged docker image
$ docker run --rm greeter:test cat greetings.txt
```

Let us add a label to our `Dockerfile` to indicate who the author and maintainer is:

```
FROM python:3.14-slim-trixie
LABEL org.opencontainers.image.authors="martin.rippin@helse-bergen.no"
RUN echo "Hello world!" > greetings.txt
```

And then we build and inspect our docker image:

```
# build tagged docker image
$ docker build . -t greeter:test
# run tagged docker image
$ docker inspect greeter:test
```

Next, we are setting `cat greetings.txt` as a default command that is run whenever the container is started:

```
FROM python:3.14-slim-trixie
LABEL org.opencontainers.image.authors="martin.rippin@helse-bergen.no"
RUN echo "Hello world!" > greetings.txt
CMD ["cat", "greetings.txt"]
```

And then we build and run our docker image:

```
# build tagged docker image
$ docker build . -t greeter:test
# run tagged docker image
$ docker run --rm greeter:test
```

There is a small python application in this repository that we would like to include in our docker image:

```
FROM python:3.14-slim-trixie
LABEL org.opencontainers.image.authors="martin.rippin@helse-bergen.no"
WORKDIR /usr/src/greeter
COPY pyproject.toml ./
COPY src/ ./src/
RUN pip install --no-cache-dir .
CMD ["greeter"]
```

And then we build and run our docker image:

```
# build tagged docker image
$ docker build . -t greeter:test
# run tagged docker image
$ docker run --rm greeter:test
```

A full list of `Dockerfile` instructions can be found here:

<https://docs.docker.com/reference/dockerfile#overview>

Apptainer

What is it?

- container platform designed for ease-of-use on shared systems and in high performance computing (HPC) environments

How is it different from Docker?

Apptainer	Docker
without root-privileges by default	requires root privileges for most operations
SIF (Singularity Image Format)	Docker/OCI images
can run Docker/OCI images	cannot run SIF images

Let's explore! 🌐

Similar to Docker, Apptainer provides a cli:

```
# check for any apptainer container
$ apptainer instance list -a
# pull image
$ apptainer pull docker://ubuntu:26.04
# check identity
$ whoami
# check identity in container
$ apptainer exec docker://ubuntu:26.04 whoami
```

The container image is saved as a `.sif` -file to the working directory.

Building an image

We are creating a file called `greeter.def` in the root of our repository and add the following lines:

```
Bootstrap: docker
From: python:3.14-slim-trixie

%post
    echo "Hello world!" > /usr/src/greetings.txt
```

And then we build and run our apptainer image:

```
# build apptainer image
$ apptainer build greeter.sif greeter.def
# execute apptainer image
$ apptainer exec greeter.sif cat /usr/src/greetings.txt
```

A full list of apptainer definition file sections can be found here:

https://apptainer.org/docs/user/main/definition_files.html#sections

3. Continuous Integration (CI)



What is it?

- automating integration of code changes from multiple contributors into single software project
- developers frequently merge code changes into central repository where automated tools are used to assert new code's correctness before integration (test, lint, build)

Which benefits does it provide?

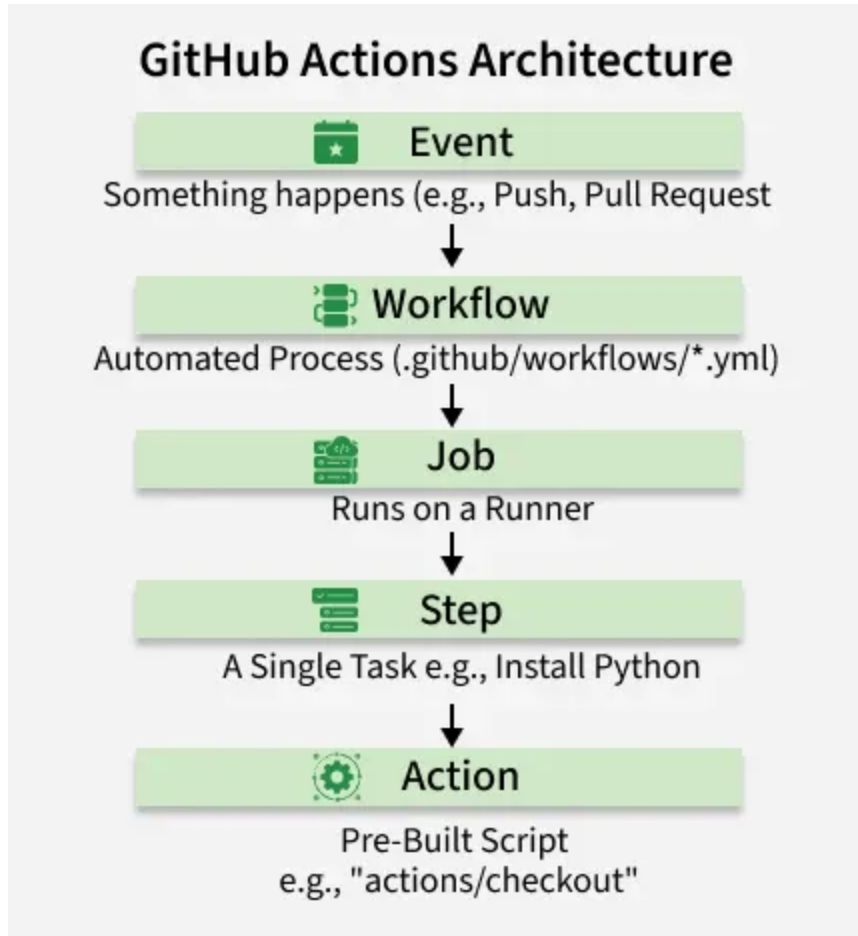
- scale up delivery output
- enables parallel work on features

GitHub actions

What is it?

- GitHub's continuous integration automation platform
- repetitive tasks are triggered based on code pushes, pull requests, or custom schedules

How does it work? 🤔



<https://www.geeksforgeeks.org/git/introduction-to-github-actions>

Workflow:

- `.yaml` -file located at `.github/workflows/` containing CI instructions

Event

- defined in workflow file
- trigger to start workflow, e.g. push, pull_request, schedule, etc.

Jobs

- defined in workflow file
- contains several tasks/*steps* and is running on a specific runner (server executing the code of the workflow)
- jobs run in parallel or can depend on each other, e.g. testing before building a container image

Step

- single task inside a job, e.g. installing python

GitHub Action

- reusable steps written by other developers that are freely available
- shorter workflows and avoiding unnecessary repetition

Example

```
name: Hello World
# event definition
on: [push]

jobs:
  # job definition
  say-hello:
    # runner definition
    runs-on: ubuntu-latest

    steps:
      # github action definition, more information about the checkout action at https://github.com/actions/checkout
      - name: Checkout Code
        uses: actions/checkout@v7

      # step definition
      - name: Run a Hello World Script
        run: |
          echo "Hello, world!"
          echo "The current repository branch is ${github.ref}"
```

Let's explore! 

We are adding a workflow file to our repository:

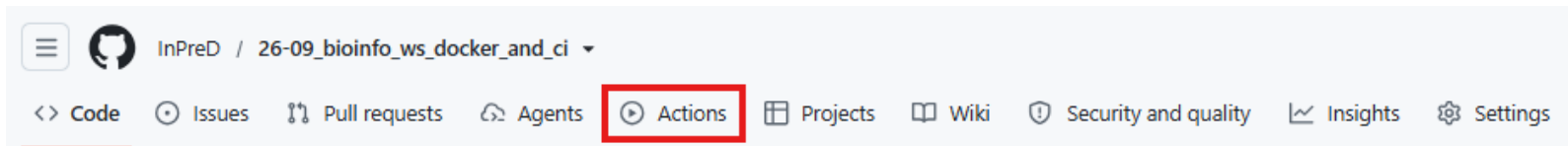
```
# create .github/workflows  
$ mkdir -p .github/workflows
```

Now we add `.github/workflows/hello_world.yaml` using the example file from the previous slide.

Now we commit our workflow doing:

```
# stage the workflow yaml file
$ git add .github/workflows/hello_world.yaml
# commit with commit message using appropriate git commit tag
$ git commit -m "ci: add hello world workflow"
# push the changes to the remote
$ git push
```

In the repository on GitHub, we navigate to **Actions**.



We see that the workflow was triggered by our last commit. To inspect we simply click on the workflow run named by our last commit message.

Actions

[New workflow](#)

All workflows

Hello World

Management

Caches

Attestations [↗](#)

Runners

Usage metrics [↗](#)

Performance metrics [↗](#)

Upcoming change to GitHub App installation token format

GitHub App installation tokens will soon use a new stateless format (ghs_...) and may be longer (~520 characters). Apps with hardcoded length assumptions may break.

Validate your apps and workflows with the per-request override header detailed in this [GitHub Changelog](#).

All workflows

Showing runs from all workflows

1 workflow run

Workflow ▾ Event ▾ Status ▾ Branch ▾ Actor ▾

ci: add hello world workflow

Hello World #3: Commit [2431e9b](#) pushed by [marrjp](#)

ci-branch

now
Queued

...

Our job `say-hello` was triggered. Select the job to inspect it.

← Hello World

✓ ci: add hello world workflow #3 Re-run all jobs ⋮

Summary

All jobs ⌵

✓ say-hello

Run details

🔄 Usage

📄 Workflow file

Triggered via push 1 minute ago	Status	Total duration	Artifacts
 marrip pushed 2431e8b c1-branch	Success	7s	—

hello_world.yaml
on: push

✓ say-hello 3s

🔄 − +

We see several steps being run which we can expand by clicking on them. But most importantly, our workflow was successfully run! 🎉

say-hello

succeeded 2 minutes ago in 3s

Search logs

🔄 ⚙️

Set up job

15

1

Current runner version: '2.337.0'

2

▶ Runner Image Provisioner

9

▶ Operating System

13

▶ Runner Image

18

▶ GITHUB_TOKEN Permissions

39

Secret source: Actions

40

Prepare workflow directory

41

Prepare all required actions

42

Getting action download info

43

Download action repository 'actions/checkout@v7' (SHA:3d3c42e5aac5ba805825da76410c181273ba90b1)

44

Complete job name: say-hello

Checkout Code

05

1

▶ Run actions/checkout@v7

17

Syncing repository: InPreB/26-09_bioinfo_ws_docker_and_ci

18

▶ Getting Git version info

22

Temporarily overriding HOME='/home/runner/work/_temp/bb1914fe-6a83-4b02-9151-8ec6d2a108f6' before making global git config changes

23

Adding repository directory to the temporary git global config as a safe directory

24

/usr/bin/git config --global --add safe.directory /home/runner/work/26-09_bioinfo_ws_docker_and_ci/26-09_bioinfo_ws_docker_and_ci

25

Deleting the contents of '/home/runner/work/26-09_bioinfo_ws_docker_and_ci/26-09_bioinfo_ws_docker_and_ci'

26

▶ Determining repository object format

27

▶ Initializing the repository

44

▶ Disabling automatic garbage collection

46

▶ Setting up auth

61

▶ Fetching the repository

69

▶ Determining the checkout info

70

/usr/bin/git sparse-checkout disable

71

/usr/bin/git config --local --unset-all extensions.worktreeconfig

72

▶ Checking out the ref

76

/usr/bin/git log -1 --format=%H

77

2431e9b1e654231e21fe2f76d830e80a7c2a73a3

Run a Hello World Script

15

1

▶ Run echo "Hello, world!"

5

Hello, world!

6

The current repository branch is refs/heads/ci-branch

Post Checkout Code

05

Complete job

05

Let's use GitHub actions to lint our Dockerfile and build the image. We start by creating `.github/workflows/docker.yaml` and add the following:

```
name: Docker Lint and Build
on: [push]

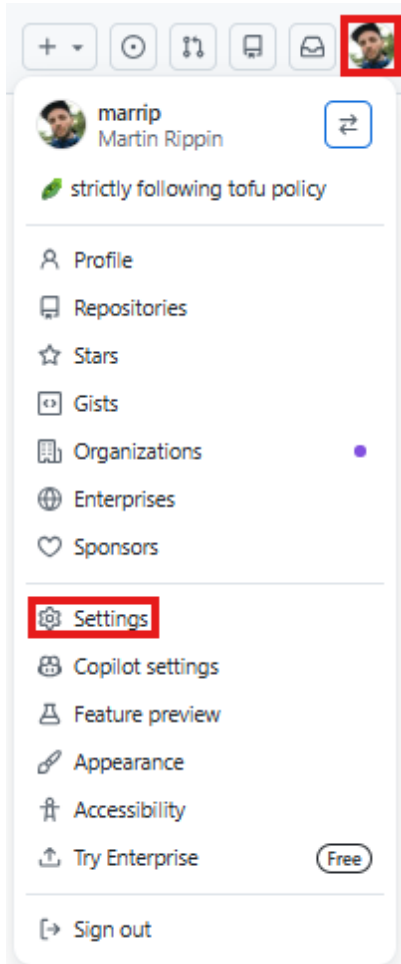
jobs:
  lint:
    name: Build Image
    runs-on: ubuntu-latest
    steps:
      # action to checkout repository
      - name: Check out the repo
        uses: actions/checkout@v7
      # action to use hadolint for linting the Dockerfile
      - name: Lint Dockerfile
        uses: hadolint/hadolint-action@v3.5.0
```

After adding the new workflow, we can commit both the `Dockerfile` and `.github/workflows/docker.yaml`:

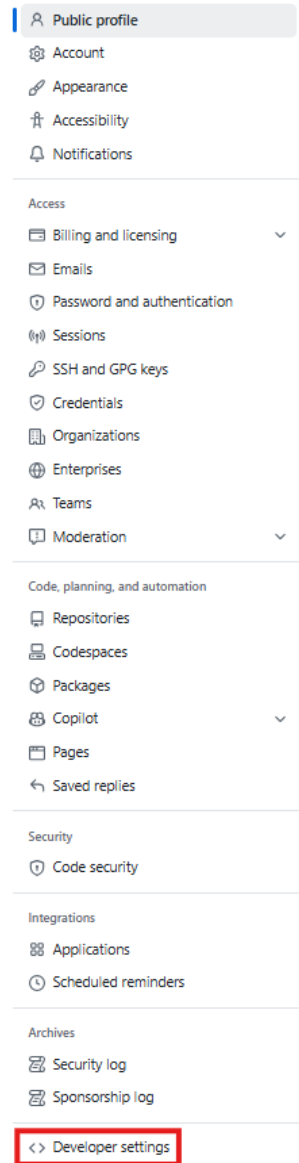
```
# stage the Dockerfile and workflow yaml file
$ git add Dockerfile .github/workflows/docker.yaml
# commit with commit message using appropriate git commit tag
$ git commit -m "ci: add Dockerfile and docker workflow"
# push the changes to the remote
$ git push
```

We navigate to `Actions` on GitHub to check if hadolint runs successfully.

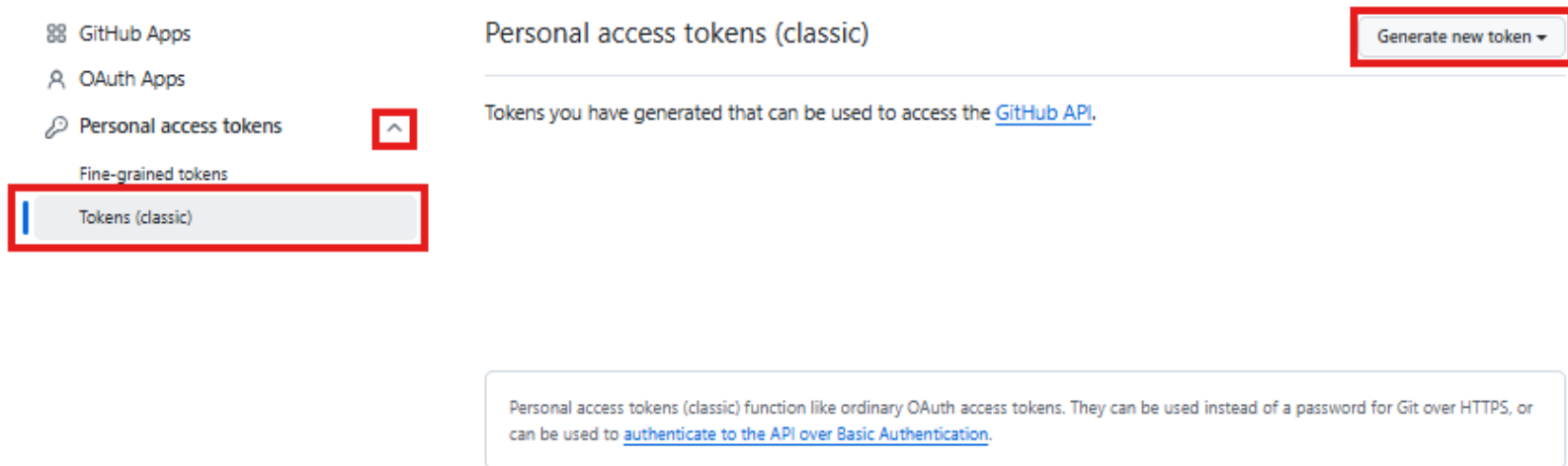
Building our docker image, we would also like to place it into the GitHub container registry (*ghcr*). To give the github action runner access to our personal registry, we need to create a personal access token (*pat*). Click on your avatar in the right corner and select **Settings** from the dropdown.



In the menu on the left select **Developer settings**.



Expand Personal access tokens and select Tokens (classic) > Generate new token > Generate new token (classic).



Give the token a descriptive name `ghcr_push_token`, select an `Expiration` (7 days should be enough) and select the `write:packages` scope (will automatically select other necessary scopes)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

ghcr_push_token

What's this token for?

Expiration

📅 7 days (Sep 16, 2026) ▾

The token will expire on the selected date

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

☒ repo	Full control of private repositories
☒ repo:status	Access commit status
☒ repo_deployment	Access deployment status
☒ public_repo	Access public repositories
☒ repo:invite	Access repository invitations
☒ security_events	Read and write security events
☐ workflow	Update GitHub Action workflows
☒ write:packages	Upload packages to GitHub Package Registry
☒ read:packages	Download packages from GitHub Package Registry
☐ delete:packages	Delete packages from GitHub Package Registry

Scroll to the bottom of the page and confirm with **Generate token**.

- ☐ **admin:ssh_signing_key** Full control of public user SSH signing keys
- ☐ **write:ssh_signing_key** Write public user SSH signing keys
- ☐ **read:ssh_signing_key** Read public user SSH signing keys

Generate token

[Cancel](#)

Copy the token.

Personal access tokens (classic)

Generate new token ▾

Tokens you have generated that can be used to access the [GitHub API](#).

 Make sure to copy your personal access token now. You won't be able to see it again!

✓ ghp_

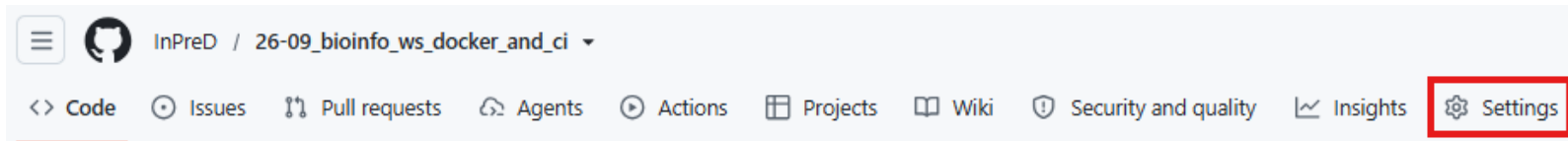


Delete

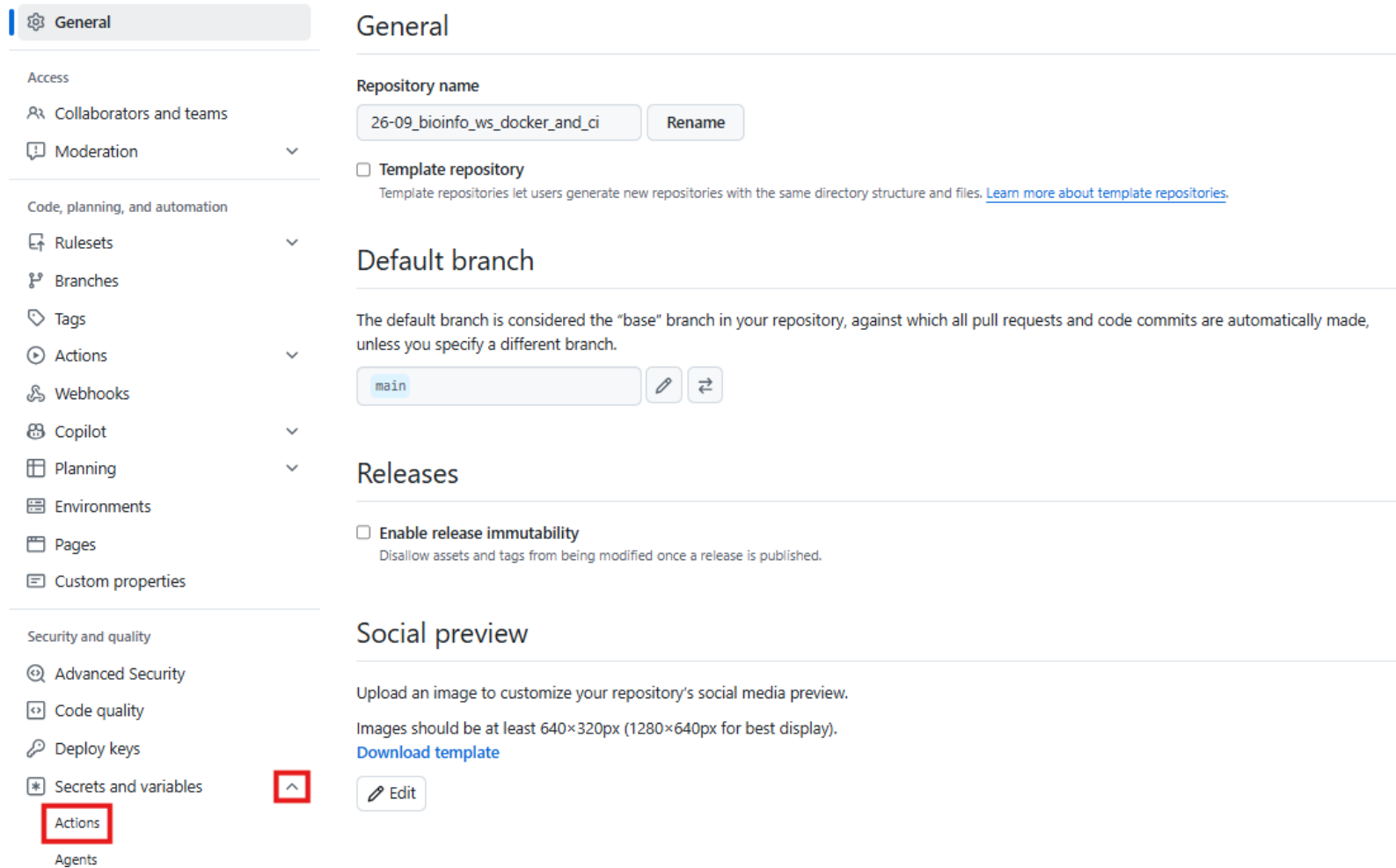
[!WARNING]

The token will only be accessible after creation, do not leave this site until you have completed adding it to your repository.

Go to your repository and click on **Settings** .



In the menu on the left, expand **Secrets and variables** and select **Actions** .



The screenshot displays the GitHub repository settings interface. On the left sidebar, the 'Secrets and variables' menu is expanded, and the 'Actions' option is highlighted with a red rectangle. A red square with an upward-pointing arrow is positioned to the right of the 'Secrets and variables' menu item. The main content area shows the 'General' settings tab. Under 'Repository name', the name '26-09_bioinfo_ws_docker_and_ci' is shown with a 'Rename' button. The 'Template repository' checkbox is unchecked. The 'Default branch' section shows 'main' as the default branch. The 'Releases' section has the 'Enable release immutability' checkbox unchecked. The 'Social preview' section includes instructions on image dimensions and a 'Download template' link. An 'Edit' button is visible at the bottom of the 'Social preview' section.

General

Repository name

26-09_bioinfo_ws_docker_and_ci **Rename**

☐ **Template repository**
Template repositories let users generate new repositories with the same directory structure and files. [Learn more about template repositories.](#)

Default branch

The default branch is considered the “base” branch in your repository, against which all pull requests and code commits are automatically made, unless you specify a different branch.

main **Edit** **↺**

Releases

☐ **Enable release immutability**
Disallow assets and tags from being modified once a release is published.

Social preview

Upload an image to customize your repository's social media preview.
Images should be at least 640×320px (1280×640px for best display).
[Download template](#)

Edit

Now we give our secret a descriptive name, basically the same as before `GHCR_PUSH_TOKEN` , add the secret we have copied (`ghp_*`) and confirm with `Add secret` .

[Actions secrets](#) / New secret

Name *

GHCR_PUSH_TOKEN

Secret *

ghp_*

Add secret

The secret should now be in your list of secrets and we can start to use it.

Now we expand our docker workflow by adding the build job below the lint job:

```
name: Docker Lint and Build
on: [push]

jobs:
  lint:
    ...
  build:
    name: Build Image
    runs-on: ubuntu-latest
    needs: lint # wait for lint to complete successfully
    steps:
      - name: Check out the repo
        uses: actions/checkout@v7
      # action to login to GitHub container registry
      - name: Login to GitHub container registry
        uses: docker/login-action@v4
        with:
          registry: ghcr.io # address to GitHub container registry
          username: ${github.actor} # the user triggering the workflow
          password: ${secrets.GHCR_PUSH_TOKEN} # The personal access token we have created earlier
      # action to build and push the image to GitHub container registry
      - name: Build and push image to GitHub container registry
        uses: docker/build-push-action@v5
        with:
          push: true
          tags: |
            ghcr.io/${github.actor}/greeter:latest
```

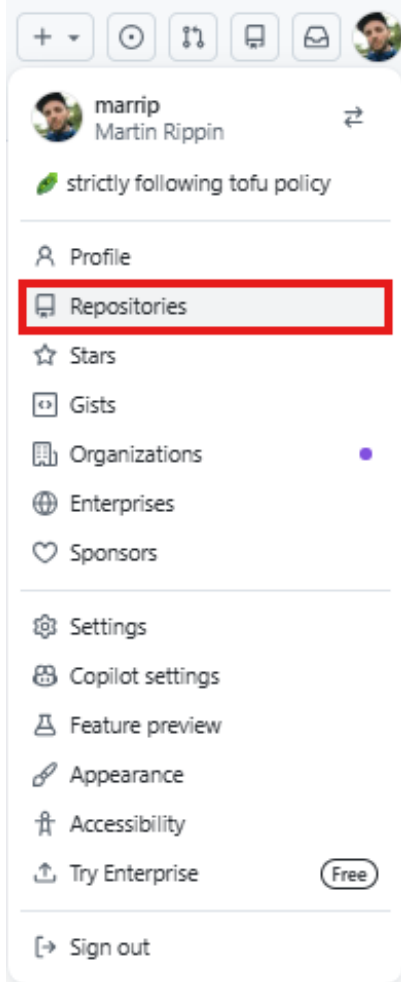
Again, we commit and push:

```
# stage the workflow yaml file
$ git add .github/workflows/docker.yaml
# commit with commit message using appropriate git commit tag
$ git commit -m "ci: add build job to docker workflow"
# push the changes to the remote
$ git push
```

And we check **Actions** to see if the workflow completes successfully.

When the image was successfully pushed to the GitHub container registry we will have to change its visibility. Click on your avatar in the right upper corner and go to

Repositories .



Continue to **Packages** .

 **marrip**

 Overview  Repositories **42**  Projects  **Packages**  Stars **124**

Click on the package `greeter` .

Type: All ▾

🔍 Search packages...

Visibility: All ▾

Sort by: Most downloads ▾

📦 1 package

📦 greeter

Private

Published 4 minutes ago by [Martin Rippin](#)

📄 1

In the menu on the right, navigate to **Package settings**.

 **greeter** latest Private Latest

 Install from the command line

[Learn more about packages](#)

```
$ docker pull ghcr.io/marrrip/greeter:latest
```

Last published

5 minutes ago

Total downloads

1

Recent tagged image versions

latest

Published 5 minutes ago · Digest 

 1

[View and manage all versions](#)

 **Package settings**

On the bottom of the page you can find `change visibility` . Click on it.

Manage Actions access



Actions repository access
Pick the repositories that can access this package using GitHub Actions.

[Add Repository](#)

[Learn more](#)

Manage Codespaces access



Codespaces repository access
Pick the repositories that can read this package using GitHub Codespaces.

[Add Repository](#)

Manage access

[Invite teams or people](#)

☐ 1 member

☐  **Martin Rippin**
marrip

Role: **Admin** - Remove

Danger Zone

Change package visibility
This package is currently private.

[Change visibility](#)

Delete this package
Once you delete a package, there is no going back. Please be certain.

[Delete this package](#)

Now we select `Public` and enter `greeter` in the text field. Finally, we confirm `I understand the consequences, change package visibility`.

Change package visibility

☒ Public

Make this package visible to anyone.

☐ Private

Make this package visible privately, to organization members that have access.

Please type greeter to confirm: *

greeter

I understand the consequences, change package visibility

This enables us now to pull our image from the GitHub container registry. Go to the command line in your repository and start the container:

```
# pull and run pushed container image  
$ docker run -it --rm ghcr.io/marrip/greeter:latest
```

If you see the greeting `Hello! Hei!` on the bottom of the standard output you have successfully pulled and run your container image. 🎉

Additional task

1. Refactor `main()` in `src/greeter/main.py` and allow the new function to greet someone by name. The name can be supplied via a prompt.
2. Add a unit test to `src/greeter/main_test.py`.
3. Add a github action job with a step to run your unit test when pushing to the repo.
4. Add a github action step to run `ruff` linting (python) when pushing to the repo.

Thank you for your attention!

Workshop completed ✓



References

- <https://www.aquasec.com/blog/a-brief-history-of-containers-from-1970s-chroot-to-docker-2016/>
- <https://ciq.com/blog/ten-years-of-apptainer-singularity-a-look-back-at-the-big-bang-of-hpc-containers>
- <https://www.freecodecamp.org/news/an-introduction-to-docker-and-containers-for-beginners/>
- <https://cyberpanel.net/blog/apptainer-vs-docker>
- <https://www.atlassian.com/continuous-delivery/continuous-integration>
- <https://www.geeksforgeeks.org/git/introduction-to-github-actions/>